CSE370 : Database Systems

Assignment 03 | Fall 2024

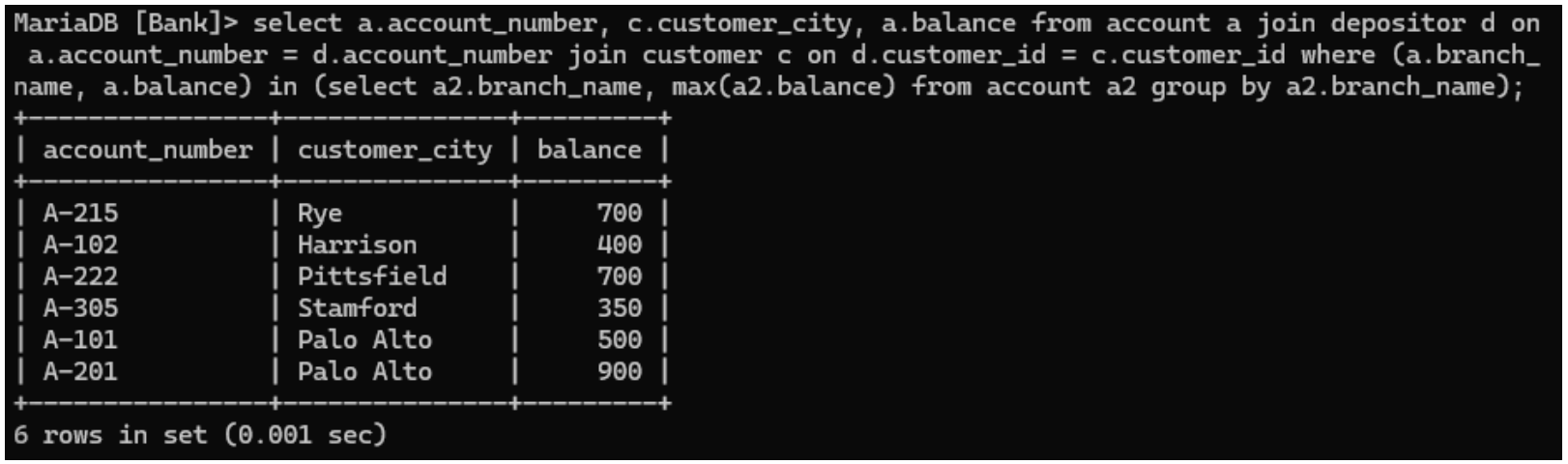
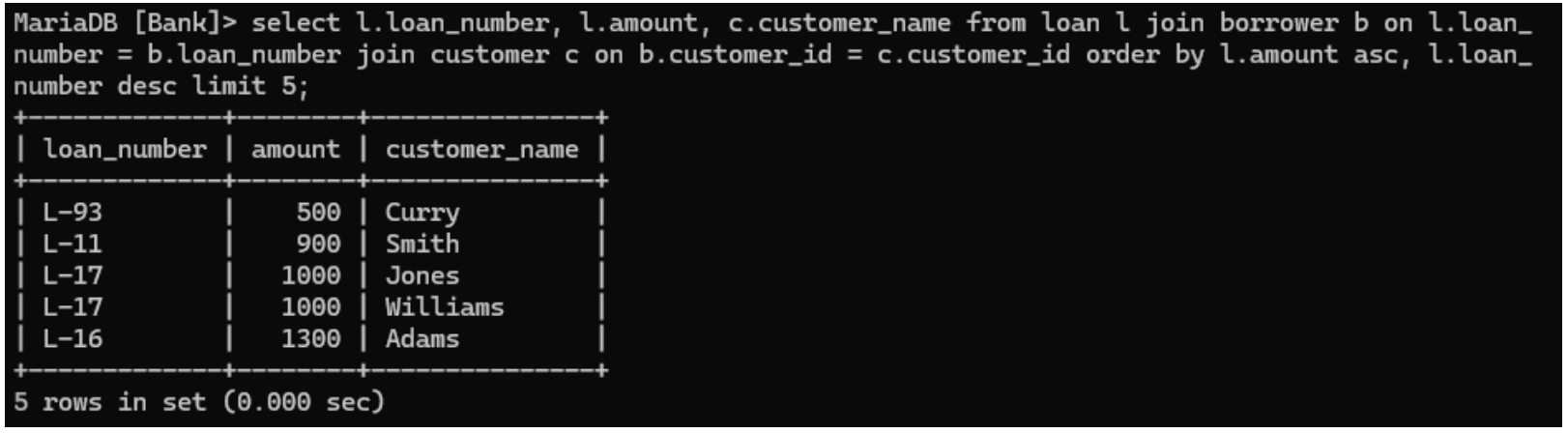
ID : 21101177 | Name : Muaz Ibne Ashraf

<table>
<tr><td>No 1 Query
(as Plain Text)</td><td>select c.customer_name, b.loan_number from customer c join borrower b on c.customer_id = b.customer_id join loan l on b.loan_number = l.loan_number where l.branch_name = 'Downtown';</td></tr>
<tr><td>No 1 SS
(of Query & Output in Shell)</td><td><pre>MariaDB [Bank]> select c.customer_name, b.loan_number from customer c join borrower b on c.customer_id
 = b.customer_id join loan l on b.loan_number = l.loan_number where l.branch_name = 'Downtown';
+---------------+-------------+
| customer_name | loan_number |
+---------------+-------------+
Johnson	L-14
Jones	L-17
Williams	L-17
+---------------+-------------+
3 rows in set (0.000 sec)</pre></td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>No 2 Query
(as Plain Text)</td><td>select c1.customer_name as Customer1, c2.customer_name as Customer2, c1.customer_city as City from customer c1 join customer c2 on c1.customer_city = c2.customer_city and c1.customer_id < c2.customer_id;</td></tr>
<tr><td>No 2 SS
(of Query & Output in Shell)</td><td><pre>MariaDB [Bank]> select c1.customer_name as Customer1, c2.customer_name as Customer2, c1.customer_city
as City from customer c1 join customer c2 on c1.customer_city = c2.customer_city and c1.customer_id <
c2.customer_id;
+-----------+-----------+------------+
| Customer1 | Customer2 | City |
+-----------+-----------+------------+
Jones	Hayes	Harrison
Smith	Curry	Rye
Lindsay	Adams	Pittsfield
Turner	Green	Stamford
+-----------+-----------+------------+
4 rows in set (0.000 sec)</pre></td></tr>
</table>

No 3 Query (as Plain Text)	select a.branch_name, sum(a.balance * 0.04) as Total_Interest from account a group by a.branch_name;

No 3 SS (of Query & Output in Shell)	`MariaDB [Bank]> select a.branch_name, sum(a.balance * 0.04) as Total_Interest from account a group by a.branch_name;`
	`+-------------+----------------+`
	`| branch_name | Total_Interest |`
	`+-------------+----------------+`
	`| Brighton | 66.00 |`
	`| Downtown | 20.00 |`
	`| Mianus | 28.00 |`
	`| Perryridge | 16.00 |`
	`| Redwood | 28.00 |`
	`| Round Hill | 14.00 |`
	`+-------------+----------------+`
	`6 rows in set (0.000 sec)`

No 4 Query (as Plain Text)	select a.account_number, c.customer_city, a.balance from account a join depositor d on a.account_number = d.account_number join customer c on d.customer_id = c.customer_id where (a.branch_name, a.balance) in (select a2.branch_name, max(a2.balance) from account a2 group by a2.branch_name);

No 4 SS (of Query & Output in Shell)	
No 5 Query (as Plain Text)	select l.loan_number, l.amount, c.customer_name from loan l join borrower b on l.loan_number = b.loan_number join customer c on b.customer_id = c.customer_id order by l.amount asc, l.loan_number desc limit 5;
No 5 SS (of Query & Output in Shell)	

No 6 Query (as Plain Text)	select distinct c.customer_name from customer c join depositor d on c.customer_id = d.customer_id join account a on d.account_number = a.account_number join borrower b on c.customer_id = b.customer_id join loan l on b.loan_number = l.loan_number where a.branch_name = 'Perryridge' and l.bra nch_name = 'Perryridge';
No 6 SS (of Query & Output in Shell)	MariaDB [Bank]> select distinct c.customer_name from customer c join depositor d on c.customer_id = d. customer_id join account a on d.account_number = a.account_number join borrower b on c.customer_id = b .customer_id join loan l on b.loan_number = l.loan_number where a.branch_name = 'Perryridge' and l.bra nch_name = 'Perryridge'; +---------------+ | customer_name | +---------------+ | Hayes | +---------------+ 1 row in set (0.000 sec)
No 7 Query (as Plain Text)	select c.customer_name, sum(l.amount) as total_loan from customer c join borrower b on c.customer_id = b.customer_id join loan l on b.loan_number = l.loan_number group by c.customer_id, c.customer_name having count(b.loan_number) >= 2;
No 7 SS (of Query & Output in Shell)	MariaDB [Bank]> select c.customer_name, sum(l.amount) as total_loan from customer c join borrower b on c.customer_id = b.customer_id join loan l on b.loan_number = l.loan_number group by c.customer_id, c. customer_name having count(b.loan_number) >= 2; +---------------+------------+ | customer_name | total_loan | +---------------+------------+ | Smith | 2900 | +---------------+------------+ 1 row in set (0.000 sec)